Piper PA-28-161 Warrior II/III (PA28-161)

Lycoming O-320-D3G, 160 hp · 4 seats · Typical Warrior II/III (1977+) POH procedures; carbureted engine, manual flaps, 2,440 lb MTOW. Warrior III figures identical unless noted.

V-speeds & limits

Vr	**50–60 KIAS (45–55 KIAS short field, flaps 25°)**
Vx	**63 KIAS**
Vy	**79 KIAS**
Best glide	**73 KIAS**
Vfe	**103 KIAS**
Va	**111 KIAS (2,440 lb) / 88 KIAS (1,531 lb)**
Vno	**126 KIAS**
Vne	**160 KIAS**
Vs / Vso	**50 / 44 KIAS**
Approach	**63 KIAS (flaps 40°)**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
Fuel quantity	**CHECK visually (25 gal usable per tank, 48 total)**
Baggage door	**CLOSED & LOCKED**
Passenger brief	**COMPLETE (belts, door, exits, sterile cockpit)**
Seats, belts, harnesses	**ADJUST & LOCK**
Door	**CLOSED & LATCHED (top and side)**
Parking brake	**SET**
Avionics master	**OFF**
Electrical switches	**OFF**
Circuit breakers	**CHECK IN**
Alternate static source	**OFF (closed)**
Fuel selector	**FULLER TANK**
Carb heat	**OFF (full forward)**

Engine Start

Mixture	**FULL RICH**
Primer	**PRIME 0–3 strokes (as required), LOCKED**
Throttle	**OPEN 1/4 inch**
Master switch	**ON**
Electric fuel pump	**ON**
Beacon	**ON**
Propeller area	**CLEAR**
Starter	**ENGAGE (10 s max, then 30 s cool-down)**
Throttle	**ADJUST 800–1,000 RPM**
Oil pressure	**CHECK rising within 30 s (60–90 psi green)**
Ammeter / alternator	**CHECK charging**
Electric fuel pump	**OFF (fuel pressure check 0.5–8 psi)**
Avionics master	**ON**
Flaps	**RETRACT (check handle latches)**

Before Taxi

Flight instruments / avionics	**CHECK & SET**
Altimeter	**SET**
Heading indicator	**SET to compass**
Transponder	**STBY / as required**
Parking brake	**RELEASE**
Brakes	**CHECK during initial roll**
Steering	**CHECK (nosewheel steering with rudder pedals)**
Gyros	**CHECK during turns (turn coordinator, DG, attitude)**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Fuel selector	**FULLER TANK**
Mixture	**FULL RICH**
Throttle	**2,000 RPM**
Magnetos	**CHECK (max drop 175 RPM, max difference 50 RPM)**
Carb heat	**CHECK (RPM drop), then OFF**
Vacuum	**4.8–5.2 in Hg**
Oil pressure / temperature	**CHECK GREEN (60–90 psi)**
Ammeter	**CHECK charging**
Annunciator panel	**PRESS TO TEST (if fitted)**
Throttle	**IDLE check (500–700 RPM), then 1,000 RPM**
Electric fuel pump	**ON**
Primer	**LOCKED**
Flaps	**SET (0° normal / 25° short field)**
Trim	**SET for takeoff (neutral)**
Seat belts / harnesses	**FASTENED**
Door	**LATCHED (top and side)**
Pitot heat / lights	**AS REQUIRED**
Transponder	**ALT**
Takeoff briefing	**COMPLETE (abort point, EFATO plan)**

After Takeoff / Climb

Airspeed	**79 KIAS (Vy) / 63 KIAS (Vx, obstacles)**
Flaps	**RETRACT (if used, when clear of obstacles)**
Electric fuel pump	**OFF (at safe altitude, check fuel pressure)**
Power	**FULL THROTTLE (climb)**
Mixture	**FULL RICH (lean above 5,000 ft)**
Engine instruments	**CHECK**
Landing light	**OFF (as desired)**

Cruise

Power	**SET (2,300–2,600 RPM, 75% max)**
Mixture	**LEAN (per power chart)**
Fuel selector	**CHANGE TANKS every 30–60 min (pump ON during change)**
Engine instruments	**CHECK**
Fuel quantity	**CHECK against plan**

Descent

Altimeter	**SET**
Mixture	**ENRICHEN as altitude decreases**
Carb heat	**AS REQUIRED (on for low power)**
Power	**REDUCE gradually (avoid shock cooling)**
Fuel selector	**FULLER TANK**
Approach briefing	**COMPLETE**

Before Landing

Seat belts / harnesses	**FASTENED**
Fuel selector	**FULLER TANK**
Electric fuel pump	**ON**
Mixture	**FULL RICH**
Carb heat	**AS REQUIRED**
Flaps	**SET (below 103 KIAS; 40° normal landing)**
Airspeed	**63 KIAS final (flaps 40°)**
Landing light	**ON**

After Landing

Flaps	**RETRACT**
Electric fuel pump	**OFF**
Carb heat	**OFF**
Transponder	**STBY / as required**
Lights	**AS REQUIRED**

Shutdown

Parking brake	**SET**
Avionics master	**OFF**
Electrical switches	**OFF**
Throttle	**1,000 RPM**
Mixture	**IDLE CUT-OFF**
Magnetos	**OFF (after prop stops)**
Master switch	**OFF**
Fuel selector	**OFF (if parking overnight / as per operator)**
Control lock / chocks	**INSTALL**

Abnormal procedures

Alternator Failure

Ammeter zero / discharge, ALT annunciator

Ammeter	**CHECK (zero output)**
ALT circuit breaker	**CHECK, RESET once**
ALT switch	**OFF, then ON**

▶ Alternator restored?

Yes:

Ammeter	**MONITOR**

No:

ALT switch	**OFF**
Electrical load	**REDUCE to minimum (one radio, transponder)**
Battery	**MONITOR (approx 30 min endurance)**
Deferred: Alternator failed – on battery power, limited electrical endurance. Land as soon as practical; no electrical flap impact (manual flaps).	**NOTED**

Low Oil Pressure

Indication below 25 psi at idle / below 60 psi in flight

Oil pressure / temperature	**CROSS-CHECK**

▶ Oil temperature rising with low pressure?

Yes – real:

Power	**REDUCE to minimum required**
Landing site	**SELECT, prepare for engine failure**
Deferred: Low oil pressure – land as soon as possible, expect engine failure.	**NOTED**

No – gauge suspect:

Flight	**CONTINUE to nearest suitable airport, monitor**
Deferred: Oil pressure indication unreliable – land as soon as practical.	**NOTED**

High Oil Temperature

Indication above 245°F / in the red

Mixture	**ENRICHEN**
Power	**REDUCE**
Airspeed	**INCREASE (lower nose)**
Oil pressure	**CHECK**

▶ Temperature decreasing?

Yes:

Flight	**CONTINUE, monitor**

No:

Deferred: Engine overheat – land as soon as possible, expect possible engine failure.	**NOTED**

Rough Running / Carburetor Ice

RPM loss, rough engine

Carb heat ..... **ON (expect initial further roughness)**

Mixture ..... **ADJUST for smoothest running**

Electric fuel pump ..... **ON**

Fuel selector ..... **SWITCH TANKS**

Magnetos ..... **CHECK L / R / BOTH (operate on best if one is faulty)**

Primer ..... **LOCKED**

▶ Engine smooth again?

Yes:

Carb heat ..... **LEAVE ON if icing conditions**

Deferred: Carb icing conditions – carb heat on at reduced power, reduced performance. ..... **NOTED**

No:

Power ..... **REDUCE to smoothest setting**

Deferred: Rough-running engine – land as soon as possible; expect engine failure. ..... **NOTED**

Low Fuel Pressure

Fuel pressure below 0.5 psi or fluctuating

Electric fuel pump ..... **ON**

Fuel selector ..... **SWITCH to fuller tank**

Mixture ..... **FULL RICH**

▶ Pressure restored?

Yes:

Electric fuel pump ..... **LEAVE ON**

Deferred: Electric fuel pump on continuously – engine-driven pump suspect. Land as soon as practical. ..... **NOTED**

No:

Fuel exhaustion ..... **CONSIDER**

Prepare for engine failure – see Engine Failure in Flight.

Flap Handle Will Not Latch / Flaps Inoperative

Flap handle ..... **RETRY, hold then release latch slowly**

▶ Flaps hold a setting?

Yes:

Flaps ..... **USE AVAILABLE setting**

No:

Flaps ..... **UP (handle stowed)**

Approach ..... **70–75 KIAS, flaps-up landing**

Deferred: Flaps inoperative – flaps-up landing, longer landing distance. ..... **NOTED**

Door Open in Flight

Airspeed ..... **SLOW to approx 89 KIAS**

Storm window ..... **OPEN (cabin vent / overhead vent closed)**

Door ..... **PULL CLOSED, latch side then top**

▶ Door latched?

Yes:

Flight ..... **CONTINUE**

No:

Aircraft is controllable with door unlatched. Expect noise and some yaw.

Deferred: Door unlatched – land normally, secure door on the ground. ..... **NOTED**

Vacuum Failure

Vacuum below 4.8 in Hg / gyros drifting

Vacuum gauge ..... **CHECK**

Attitude indicator / DG ..... **CROSS-CHECK against turn coordinator and compass**

Attitude indicator / DG ..... **COVER or disregard**

IMC ..... **EXIT, request no-gyro assistance if needed**

Deferred: Vacuum failed – no attitude indicator / DG. Avoid IMC; partial-panel landing. ..... **NOTED**

Static Source Blocked

Airspeed / altimeter / VSI erratic

Pitot heat ..... **ON**

Alternate static source ..... **OPEN (valve under left side of panel)**

Storm window / vents ..... **CLOSED (for calibration accuracy)**

Airspeed and altimeter read slightly high with alternate static; see POH correction table.

Deferred: Alternate static in use – small airspeed/altimeter errors on approach. ..... **NOTED**

Inadvertent Icing

Pitot heat	**ON**
Carb heat	**ON**
Cabin heat / defroster	**ON (windshield)**
Exit icing conditions	**TURN / CLIMB / DESCEND**
Airspeed	**INCREASE, avoid abrupt maneuvers**
Flaps	**LIMIT (flaps-up or 10° landing with residual ice)**

Deferred: Residual airframe ice – flaps-up / 10° landing at increased speed, longer runway. **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

Loss of power before rotation

Throttle (memory)	**CLOSED**
Brakes (memory)	**APPLY, stop straight ahead**
Mixture	**IDLE CUT-OFF**
Magnetos	**OFF**
Master switch	**OFF**
Fuel selector	**OFF**

Engine Failure After Takeoff

Airborne, insufficient runway remaining

LAND IMMEDIATELY

Airspeed (memory)	**73 KIAS (best glide)**
Landing area (memory)	**SELECT straight ahead (small heading changes only)**

▶ Altitude and time available for a restart attempt?

Yes:

Fuel selector	**SWITCH to other tank**
Electric fuel pump	**CHECK ON**
Mixture	**FULL RICH**
Carb heat	**ON**
Primer	**LOCKED**

No – commit to landing:

Flaps	**AS REQUIRED**
Fuel selector	**OFF**
Mixture	**IDLE CUT-OFF**
Magnetos	**OFF**
Master switch	**OFF (before touchdown)**
Door	**UNLATCH before touchdown**

Do not attempt to turn back to the runway at low altitude.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	73 KIAS (best glide)
Fuel selector (memory)	SWITCH to tank containing fuel
Electric fuel pump (memory)	ON
Mixture (memory)	FULL RICH
Carb heat (memory)	ON
Engine gauges	CHECK for cause (fuel pressure, oil, RPM)
Primer	LOCKED

▶ Propeller stopped (not windmilling)?

Yes – crank:

Magnetos	BOTH
Throttle	OPEN 1/4 inch
Starter	ENGAGE

No – windmilling:

Magnetos	CHECK BOTH (try L / R individually)
Throttle	CYCLE, then set as required

▶ Engine restarted?

Yes:

Power	REDUCE to minimum required
Engine instruments	MONITOR
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect cause (fuel/carb ice).	NOTED

No – forced landing:

Forced landing without power:

Landing area	SELECT, plan approach into wind
Transponder / radio	7700, MAYDAY (position, intentions)
Seat belts / harnesses	SECURE
Flaps	AS REQUIRED (full on final)
Airspeed	63 KIAS final (flaps 40°)
Fuel selector	OFF
Mixture	IDLE CUT-OFF
Magnetos	OFF
Master switch	OFF (when no longer required)
Door	UNLATCH before touchdown

Engine Fire in Flight

LAND IMMEDIATELY

Fuel selector (memory)	OFF
Throttle (memory)	CLOSED
Mixture (memory)	IDLE CUT-OFF
Electric fuel pump (memory)	CHECK OFF
Cabin heat / defroster (memory)	OFF
Airspeed	INCREASE to blow out fire (do not exceed 126 KIAS)
Forced landing	EXECUTE (do not attempt restart)
Master switch	OFF before touchdown

Engine Fire During Start

Starter (memory)	CONTINUE CRANKING (draw fire into engine)
Mixture (memory)	IDLE CUT-OFF
Throttle (memory)	FULL OPEN
Electric fuel pump	OFF
Fuel selector	OFF

▶ Fire continues?

Yes:

Magnetos / master	OFF
Aircraft	EVACUATE, fire extinguisher

No:

Engine	SHUT DOWN, inspect before further flight

Electrical / Cabin Fire

LAND AS SOON AS POSSIBLE

- Master switch (memory) ..... **OFF**
- Avionics master (memory) ..... **OFF**
- Cabin heat / defroster (memory) ..... **OFF**
- Vents ..... **OPEN**
- Fire extinguisher ..... **USE (if fire in cabin)**

After discharge, ventilate the cabin before landing.

▶ Fire out and electrical power essential?

Yes:

- Circuit breakers ..... **CHECK, do not reset tripped breakers**
- Master switch ..... **ON**
- Electrical items ..... **ON one at a time, watch for smoke**
- Deferred: Electrical fire – faulted circuit isolated, affected equipment unavailable. No flaps impact (manual flaps). ..... **NOTED**

No:

Continue with master off. Manual flaps and engine are unaffected; no radio, lights or transponder.

Emergency Descent

- Throttle (memory) ..... **CLOSED**
- Mixture (memory) ..... **FULL RICH**
- Carb heat (memory) ..... **ON**
- Flaps (memory) ..... **UP**
- Airspeed (memory) ..... **126 KIAS smooth air / 111 KIAS turbulent**

Reduce speed before extending flaps (103 KIAS max).

Spin Recovery

Unintentional spin

- Throttle (memory) ..... **IDLE**
- Ailerons (memory) ..... **NEUTRAL**
- Rudder (memory) ..... **FULL OPPOSITE to rotation**
- Control wheel (memory) ..... **FULL FORWARD**
- Rudder (memory) ..... **NEUTRAL when rotation stops**
- Control wheel ..... **EASE BACK, recover from dive**

Intentional spins are prohibited.

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.7 (build 849ad9ab) · 2026-11-07